Bayes classification

Preliminaries

Normal density

The Gaussian density for a single variable is given by

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{1}$$

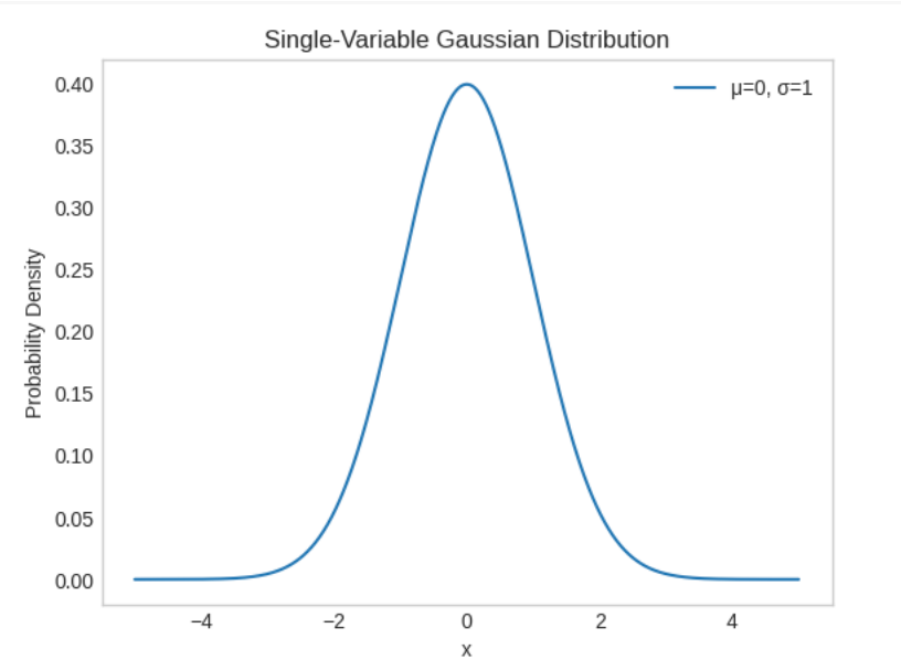

Figure 1: Univariate Gaussian density

The multivariate Gaussian density of a d-dimensional variable $\mathbf{x} \in \mathbb{R}^d$ is given by

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} exp(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)) \tag{2}$$

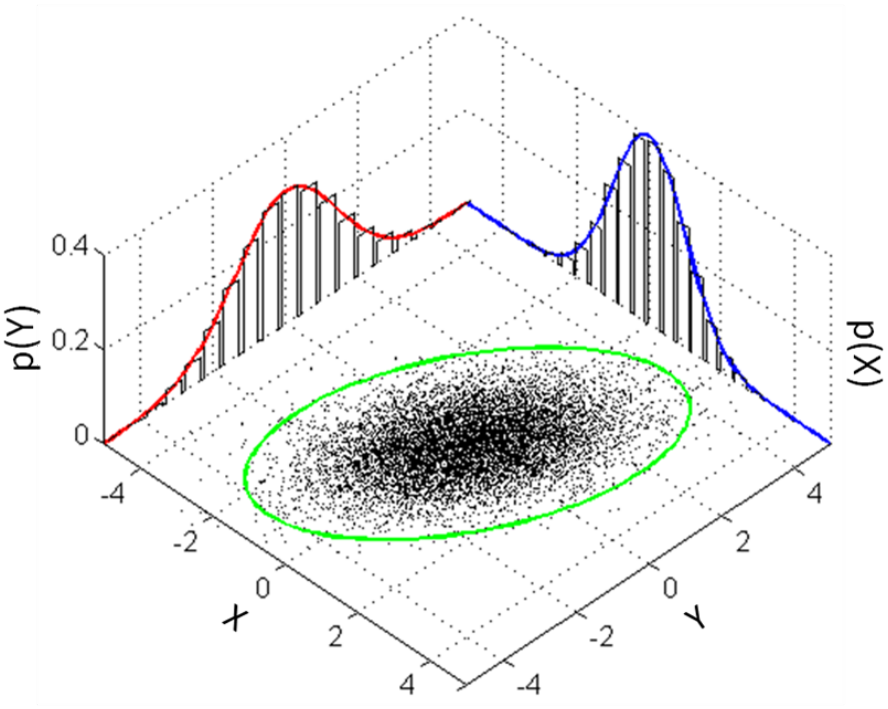

Figure 2: Multivariate Gaussian density

Bayesian Decision Theory

Bayesian Decision Theory

Goal: Minimize expected risk.

Posterior probability:

$$P(C_k|\mathbf{x}) = \frac{p(\mathbf{x}|C_k)P(C_k)}{p(\mathbf{x})}$$

Decision rule:

$$\text{Choose } C_k \text{ with maximum } P(C_k|\mathbf{x})$$

This is called the **Bayes classifier**.

Loss and Risk

Define loss function λ_{ij}:

$$\lambda_{ij} = \text{Loss of deciding } C_i \text{ when true class is } C_j$$

Expected risk:

$$R(C_i|\mathbf{x}) = \sum_j \lambda_{ij} P(C_j|\mathbf{x})$$

Choose class with minimum conditional risk.

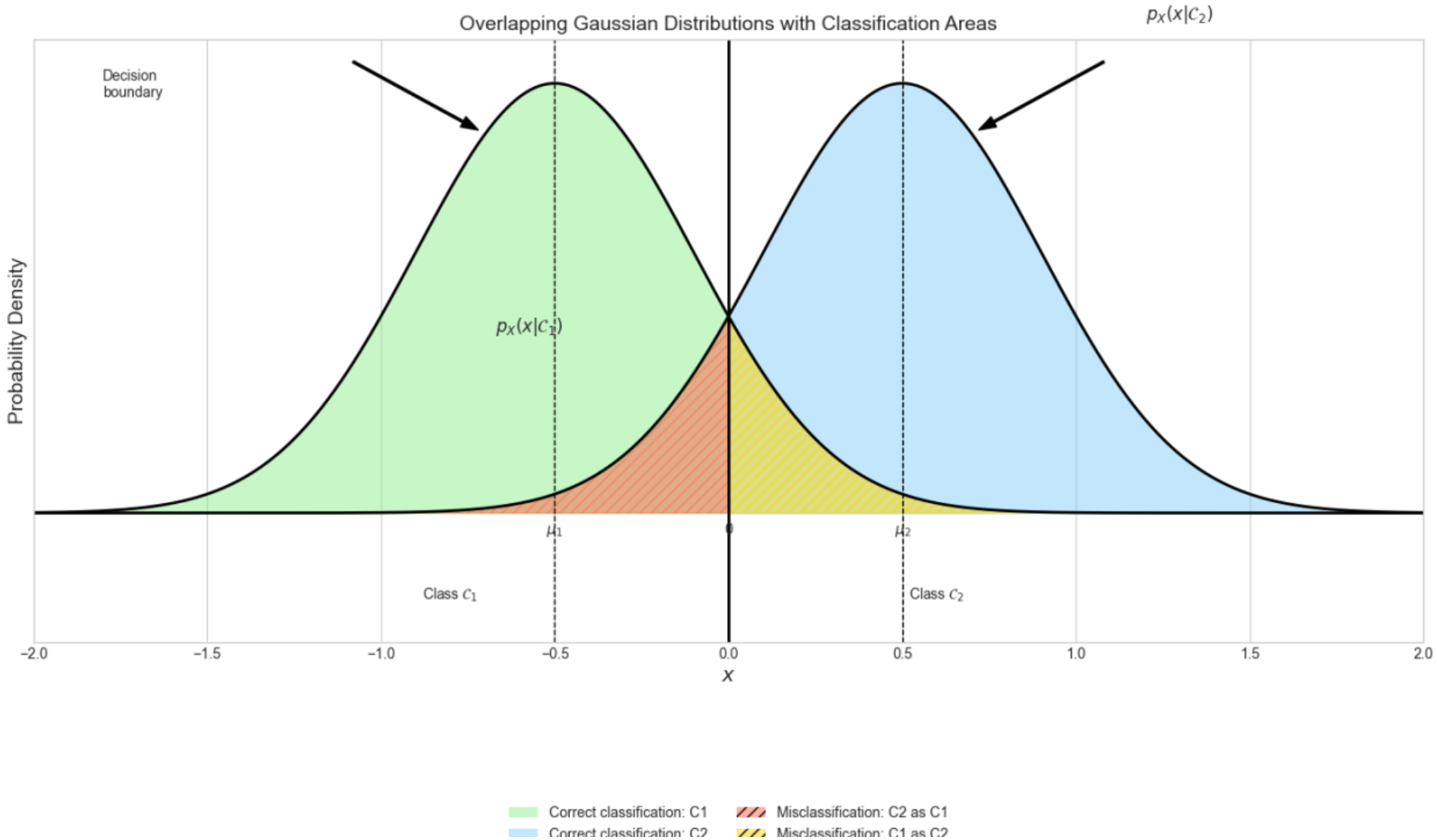

Figure 3: Two classes modeled as univariate Gaussians

Probabilistic Discriminative Classifiers

Discriminative vs Generative models

Two common types of classification models are **discriminative** and **generative** models.

- **Discriminative models:**
  - Compute a decision boundary directly (usually under some assumptions), for example directly model $P(C_k|\mathbf{x})$
  - Example : Logistic Regression.

- **Generative models:**
  - Model $p(\mathbf{x}|C_k)$ and $P(C_k)$
  - Example: Gaussian classifier

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Discriminant Functions

What is a Discriminant Function?

A discriminant function assigns a score to each class.

For class C_k:

$$g_k(\mathbf{x})$$

Decision rule:

$$\text{Assign } \mathbf{x} \to C_i \quad \text{if} \quad g_i(\mathbf{x}) > g_j(\mathbf{x}), \forall j \neq i$$

Goal: Separate classes in feature space.

Linear Discriminant Function

Linear form:

$$g_k(\mathbf{x}) = \mathbf{w}_k^T \mathbf{x} + w_{k0}$$

Decision boundary between C_1 and C_2:

$$\mathbf{w}^T \mathbf{x} + w_0 = 0$$

Properties:

- Forms a hyperplane
- Computationally efficient
- Works well for linearly separable data

Probabilistic Discriminant Function

For two classes:

$$P(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x})$$

where sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Decision rule:

$$\text{Assign } C_1 \text{ if } P(C_1|\mathbf{x}) > 0.5$$